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TVL12M

Tirunelveli District
Common Quarterly Examination - 2025

-1252

Standard 12
MATHEMATICS
PART - A

Time: 3.00 Hours Marks: 90

Choose the correct answer: **20x1=20**

1) If adj $A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ and adj $B = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ then adj (AB) is

a) $\begin{bmatrix} -7 & -1 \\ 7 & -9 \end{bmatrix}$ b) $\begin{bmatrix} -6 & -2 \\ 5 & -10 \end{bmatrix}$ c) $\begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix}$ d) $\begin{bmatrix} -7 & 7 \\ -1 & -9 \end{bmatrix}$

2) If $A = \begin{bmatrix} 3/5 & 4/5 \\ x & 3/5 \end{bmatrix}$ and $A^T = A^{-1}$, then the value of x is

a) $4/5$ b) $3/5$ c) $-4/5$ d) $-3/5$

3) The principal argument of $\frac{3}{-1+i}$ is

a) $-3\pi/4$ b) $-2\pi/3$ c) $-5\pi/6$ d) $-\pi/2$

4) If $\left|z - 3/z\right| = 2$, then the least value of |z| is

a) 4 b) 2 c) 1 d) 3

5) The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k staisfies

a) $|k| \le 6$ b) $k = 0$ c) $|k| \ge 6$ d) $|k| > 6$

6) A polynomial equation in x of degree n always has

a) n distinct roots b) n imaginary roots
c) n real roots d) atmost one root

7) If $\sin^{-1}x + \sin^{-1}y = 2\pi/3$, then $\cos^{-1}x + \cos^{-1}y =$

a) $2\pi/3$ b) $\pi/6$ c) π d) $\pi/3$

8) If $x = 1/5$, the value of $\cos[\cos^{-1}x + 2\sin^{-1}x]$ is

a) $-\sqrt{\frac{24}{25}}$ b) $\sqrt{\frac{24}{25}}$ c) $-1/5$ d) $1/5$

9) If $\sin^{-1}\frac{x}{5} + \operatorname{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is

a) 3 b) 2 c) 4 d) 5

10) The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{11}$ c) $\sqrt{10}$

11) If x + y = K is a normal to the parabola $y^2 = 12x$, then the value of K is

a) 9 b) 3 c) 1 d) -1

12) The area of quadrilateral formed with foci of the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is

a) $4(a^2 + b^2)$ b) $2(a^2 + b^2)$ c) $a^2 + b^2$ d) $\frac{1}{2}(a^2 + b^2)$

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13) If a vector α lies in the plane of β and γ, then

a) $[\alpha\ \beta\ \gamma]=1$ b) $[\alpha\ \beta\ \gamma]=0$ c) $[\alpha\ \beta\ \gamma]=-1$ d) $[\alpha\ \beta\ \gamma]=2$

14) The angle between the line $r=(\hat{i}+2j-3k)+t(2\hat{i}+j-2k)$ and the plane $r.(\hat{i}+j)+4=0$ is

a) 0^0 b) 45^0 c) 90^0 d) 30^0

15) If a,b,c are non-coplanar, non-zero vectors such that $[a,b,c]=3$, then $\{[a\times b, b\times c, c\times a]\}^2 =$

a) 18 b) 81 c) 27 d) 9

16) If the sum of the roots of the equation $(a+1)x^2+(2a+3)x+3a+4=0$ is -3 then the product of the roots is

a) 1 b) 4 c) 2 d) -2

17) A value of x statisfying $\tan^{-1}(x+3)-\tan^{-1}(x-3)=\sin^{-1}(3/5)$ is

a) 0 b) 2 c) 4 d) 8

18) Equation of directrix of the ellipse $\frac{x^2}{36}+\frac{y^2}{4}=1$ is

a) $\sqrt{2}x-9=0$ b) $\sqrt{2}x+9=0$ c) $\sqrt{3}x+9=0$ d) $\sqrt{3}x-9=0$

19) Distance between the parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is

a) $7/2$ b) $5/2$ c) $3/2$ d) $9/2$

20) Which of the following is correct?

a) $z=\bar{z}$ then z is imaginary b) $z=-\bar{z}$ then z is real

c) $|z|=z\bar{z}$ d) $\mathrm{Re}(z)=\frac{z+\bar{z}}{2}$

PART - B

Answer any 7 questions. Q No. 30 is compulsory: **7x2=14**

21) If $\text{adj}A=\begin{bmatrix}-1 & 2 & 2\\ 1 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, find A^{-1}

22) If $A=\begin{bmatrix}2 & 9\\ 1 & 7\end{bmatrix}$ find $\left[A^{-1}\right]^T$

23) Write in polar form: $-1-i$

24) Solve : $x^4-14x^2+45=0$

25) Find the principal value of $\tan^{-1}(-\sqrt{3})$

26) If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\pi$, show that $x+y+z=xyz$.

27) Find the centre and radius of the circle $3x^2+(a+1)y^2+6x-9y+a+4=0$.

28) A partice is acted upon by the forces $3i-2j+2k$ and $2i+j-k$ is displaced from the point (1, 3, −1) to the point (4, −1, λ). If the work done by the forces is 16 units, find the value of λ.

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29) Show that the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{3}$ and $\frac{x-1}{-3}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar

30) If e_1 is the eccentricity of the hyperbola $\frac{x^2}{49}-\frac{y^2}{36}=1$ and e_2 is the eccentricity of the hyperbola $\frac{x^2}{36}-\frac{y^2}{49}=1$ then prove that $\frac{1}{e_1^2}+\frac{1}{e_2^2}=1$

PART - C

Answer any 7 questions. Q.No. 40 is compulsory: **7x3=21**

31) Solve: $2x + 2y + z = 5$, $x - y + z = 1$, $3x + y + 2z = 4$ by using rank method.

32) Show that $\left(2+i\sqrt{3}\right)^{10}-\left(2-i\sqrt{3}\right)^{10}$ is purely imaginary.

33) If $x=\cos\alpha+i\sin\alpha$, $y=\cos\beta+i\sin\beta$ then show that $x^m y^n+\frac{1}{x^m y^n}=2\cos\left[m\alpha+n\beta\right]$

34) Solve the equation $x^3 - 6x^2 - 4x + 24 = 0$ roots are in arithmetic progression

35) Find the value of $\cot^{-1}(1)+\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)-\sec^{-1}\left(-\sqrt{2}\right)$

36) Solve : $2\tan^{-1}x=\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)+\cos^{-1}\left(\frac{1-b^2}{1+b^2}\right)$ a, b>0

37) Find the equation of the ellipse if the length of latus-recturm is 4 and distance between foci $4\sqrt{2}$ and major axis as y-axis

38) Find the equation of tangent and normal to the parabola $x^2 + 6x + 4y + 5 = 0$ at (1, −3).

39) Find the shortest distance between the two given straight lines $r=\left(2i+3j+4k\right)+t\left(-2i+j-2k\right)$ and $\frac{x-3}{2}=\frac{y}{-1}=\frac{Z+2}{2}$

40) Find the angle between the lines $\frac{x-2}{3}=\frac{y+1}{-2}, z=2$ and $\frac{x-1}{1}=\frac{2y+3}{3}=\frac{z+5}{2}$

PART - D

Answer all the questions: **7x5=35**

41) Solve: $3x + 3y - z = 11$
$2x - y + 2z = 9$
$4x + 3y + 2z = 25$
by using Cramer's rule.

(OR)

Evaluate: $\sin\left[\sin^{-1}\left(\frac{3}{5}\right)+\sec^{-1}\left(\frac{5}{4}\right)\right]$

42) Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point (2, 3, 6) and parallel to the straight lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-3}{1}$ and $\frac{x+3}{2}=\frac{y-3}{-5}=\frac{z+1}{-3}$

(OR)

If z_1, z_2 and z_3 are three complex numbers such that $|z_1|=1$, $|z_2|=2$, $|z_3|=3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2+4z_1z_3+z_2z_3| = 6$

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43) Find the centre, foci and eccentricity of the hyperbola
$11x^2-25y^2-44x+50y-256=0$

(OR)

Solve : $(x-5)(x-7)(x+6)(x+4) = 504$

44) Prove by Vector method, $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

(OR)

Solve : $\tan^{-1}\left[\frac{x-1}{x-2}\right] + \tan^{-1}\left[\frac{x+1}{x+2}\right] = \frac{\pi}{4}$

45) An equation of the elliptical part of an optical lens system is $\frac{x^2}{16}+\frac{y^2}{9}=1$.
The parabolic part of the system has a focus in common with the right focus of the ellipse. The vertex of the parabola is at the origin and the parabola opens to the right determine the equation of the ellipse

(OR)

If $z = x + iy$, and $\arg\frac{z-i}{z+2}=\frac{\pi}{4}$, show that $x^2+y^2+3x-3y+2=0$

46) If $2 + i$ and $3-\sqrt{2}$ are the roots of the equation
$x^6-13x^5+62x^4-126x^3+65x^2+127x-140 = 0$

(OR)

Determine the values of λ, for which the equation $x - y + 3z = 0$, $4x + 3y + \lambda z = 0$, $2x + y + 2z = 0$ has (i) a unique solution (ii) a non - trivial solution.

47) Find the anlge between the line $\frac{x+1}{2}=\frac{y}{3}=\frac{z-3}{6}$ and the plane $10x+2y-11z=3$

(OR)

Prove that $\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{8}\right)=\frac{\pi}{4}$

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